Multi-Omics Graph Attention Network for Key Gene Prediction in Triple-Negative Breast Cancer

Zixi Jiang, Kexin Cao, Fang Ge, and Zhusheng Huang

***Abstract*—Triple-negative breast cancer (TNBC) is an aggressive malignancy lacking effective targeted therapies, underscoring the need for robust and interpretable biomarkers for personalized treatment. Here, we propose a graph attention network (GAT)-based multimodal framework that integrates scRNA-seq, scATAC-seq, and radiomics to capture cross-modal regulatory interactions underlying TNBC heterogeneity. Transcriptional, chromatin accessibility, and imaging features are aligned via canonical correlation analysis, with intercellular communication-derived gene relationships and transcription factor-binding-guided edges incorporated into a multimodal graph. Multi-head attention enables adaptive weighting of omics-specific interactions, while an ensemble multilayer perceptron with variational dropout stratifies patient prognosis. The model demonstrates strong predictive performance in an external TCGA-TNBC cohort (log-rank p<0.01), outperforming single-omics and alternative graph-based approaches (AUC-ROC = 0.839, 95% CI: 0.81–0.87). Pathway analysis validates canonical TNBC drivers, including PI3K, ERBB2, PTK6, and EGFR signaling, while revealing previously underappreciated regulatory programs involving complement-coagulation cascades, ECM-integrin-focal adhesion signaling, leukocyte transendothelial migration, sphingolipid-mediated metabolic-immune coupling, and nanoparticle-receptor interactions. Collectively, this framework provides an interpretable strategy for multimodal biomarker discovery in TNBC, uncovering both established and novel therapeutic vulnerabilities and offering a scalable approach toward precision oncology.**

***Index Terms*—Triple-negative breast cancer (TNBC), multi-omics data integration, graph attention network (GAT), single-Cell RNA sequencing (scRNA-seq), key gene prediction, prognostic biomarkers.**

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I. Introduction

TRIPLE-NEGATIVE breast cancer (TNBC), an aggressive subtype lacking estrogen receptor (ER), progesterone receptor (PR), and HER2 expression, remains a critical unmet need in oncology due to intrinsic resistance to targeted therapies, high relapse rates, and poor prognosis [1]. Its profound molecular heterogeneity, both inter- and intratumoral, further complicates treatment, as conventional single-omics approaches fail to capture the cross-modal interactions governing tumorigenesis [2]. Recent advances in multi-omics technologies (scRNA-seq, scATAC-seq, radiomics) and biological network databases (STRING with 707236 PPIs, HGD, CellPhoneDB) have illuminated key tumor microenvironment (TME) crosstalk, such as CAF-macrophage IL6-IL6R signaling [1], but these insights remain fragmented. Existing biological research often focuses on tumor-intrinsic oncogenic pathways (e.g., PI3K-Akt, EGFR) while neglecting underappreciated regulatory layers, creating a gap in understanding multi-scale TME dynamics [2].

Biological studies of TNBC face two critical limitations: a "phenotype-molecular gap" and fragmented network analysis. Radiomic features (e.g., tumor texture heterogeneity) are linked to clinical outcomes [3], but few studies systematically map these imaging phenotypes to molecular alterations, limiting translation to personalized therapy. Additionally, while public databases reveal discrete network layers (TF regulation, cell-cell communication), current research analyzes them independently, obscuring how innate immune-vascular crosstalk, mechanotransduction, and metabolic-immune coupling jointly drive progression [1]. Notably, pathways such as complement and coagulation signaling, ECM-integrin-focal adhesion, leukocyte transendothelial migration, and sphingolipid signaling, all critical for immune regulation and TME remodeling, remain insufficiently characterized in TNBC, highlighting the need for integrative approaches to uncover systemic drivers [2].

Computational methods for multi-omics integration have evolved but still face key drawbacks. Early linear methods (CCA, MOFA+ [4]) and matrix factorization (NMF) excel at dimensionality reduction but fail to capture non-linear biological networks [5]. Graph-based models like GCN [6] enable network-aware aggregation but use uniform node weighting, limiting adaptation to context-specific gene relevance. Advanced models such as scMoGNN [7] (single-cell multi-omics GNN) and MOH

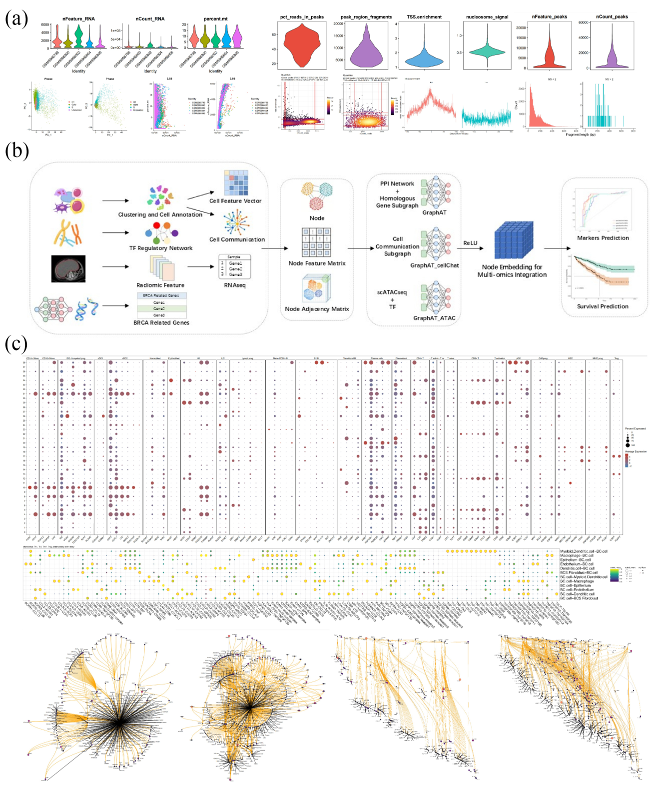

Fig. 1. Multi-Omics Integration with GAT for Key Gene Discovery and Prognostic Prediction in TNBC. (a) Dataset collection. (b) GAT to integrate transcriptomics, genomics, and radiomics data for identifying key genes in TNBC. (c) The model captures complex gene interactions and regulatory networks, uncovering novel biomarkers and predictive signatures for improved prognostic models.

[8] (heterogeneous graph integrtion) improve subtype classification but lack adaptive attention mechanisms, support for radiomic data, and biological interpretability [9]. Transformer-based methods [10] and graph contrastive learning [11] enhance feature robustness but prioritize performance over interpretability, with no feature attribution to clarify driver omics layers or edges [4]. Prognostic models often lack rigorous regularization (e.g., variational dropout) and standardized hyperparameter tuning, undermining reproducibility [4].

To address these challenges, we propose a GAT-based multimodal framework that integratively analyzes scRNA-seq, scATAC-seq, CITE-seq, and radiomics data to decode the regulatory complexity of TNBC across molecular and phenotypic scales. By explicitly modeling biologically informed interactions and leveraging attention-based learning, this framework enables robust biomarker discovery and prognostic stratification in small-sample, high-dimensional settings Fig. 1(a)–(c). Specifically, our work makes the following contributions:

1) Unlike existing GAT-based multi-omics approaches that rely on implicit modality fusion, our framework explicitly models cross-modal regulatory interactions, particularly chromatin accessibility–guided transcriptional regulation, by incorporating transcription factor binding-informed edges between scATAC-seq and scRNA-seq features.
2) We introduce a unified graph attention framework that integrates single-cell multi-omics data with radiomics, enabling multiscale characterization of TNBC biology from intracellular regulatory programs to tumor-level phenotypic manifestations, an aspect largely unexplored in prior GAT-based integration studies.
3) Our model is specifically designed for small-sample, high-dimensional multimodal scenarios, employing ensemble learning with variational dropout to enhance robustness and generalization, and demonstrating superior performance over state-of-the-art methods in low-sample benchmarks.
4) Beyond representation learning, the proposed framework prioritizes biologically interpretable driver gene identification and clinically relevant prognostic stratification, supported by attention-based mechanistic insights and independent survival validation in the TCGA-TNBC cohort.

II. Methods

A. Dataset Collection

To investigate the molecular mechanisms underlying breast can-cer (BRCA) and predict key genes, we gathered multi-omics da-tasets, including single-cell RNA sequencing (scRNA-seq), single-cell chromatin accessibility sequencing (scATAC-seq), and radi-omics data (Table S1). The Table S1 provides details of each dataset. Additionally, a search in the MalaCards database identified 1243 breast cancer-related genes, of which 208 are categorized as hub genes, with 112 of these experimentally validated, excluding hub genes.

B. Preprocessing of scRNA-seq and CITE-seq

To ensure data quality and reliability, we preprocessed the collected scRNA-seq data using the Seurat package. First, low-quality cells were removed based on the thresholds: nFeature RNA > 200 & nFeature RNA < 7500 & percent.mt< 20. For RNA sequences, we normalized the data using the NormalizeData method with normalization.method set to “LogNormalize” and scale.factor to 10000. High-variance genes were identified using the FindVariableFeatures method. Subsequently, we calculated G2M and S phase scores for each cell using the CellCycleScoring method and regressed out the cell cycle effects during normalization using ScaleData. For ADT sequences in CITE-seq, we also normalized the data using the NormalizeData method with normalization.method set to “CLR” and margin set to 2.

C. Preprocessing of scATAC-seq

To ensure data quality and analysis accuracy, we preprocessed the scATAC-seq data. First, we converted raw Fastq data to a 10X matrix format using the cellranger-atac software. Then, using the Signac and Seurat packages, we transformed it into a Seurat object with gene annotation version “hg38”. Quality control metrics were computed using the Signac package, applying thresholds (*peak region fragments* $>$ *2000 & peak region* *fragments < 20000 & pct reads in peaks > 15 & nucleosome signal < 4 & TSS.enrichment > 1*) for sample quality control.

D. Radiomic Feature Extraction

Radiomic features were extracted using the pyradiomics package. Prior to feature extraction, a total of 88 samples were included in the study. First, we segmented the images based on lesion location annotations provided with the dataset. Following segmentation, we extracted a total of 35 radiomic features from the 88 segmented images. These features were derived from the segmented regions of interest and included parameters such as shape, intensity, texture, and signal heterogeneity.

E. Dimensionality Reduction, Clustering, and Cell an Notation for scRNA-seq + CITE

For the scRNA-seq and CITE-seq data, we first performed principal component analysis (PCA) separately on both sequences to reduce dimensionality and identify major sources of variation. Clustering was conducted on the reduced data using FindMultiModalNeighbors and FindClusters methods. The Run Uniform Manifold Approximation and Projection (RunUMAP) method was then employed to calculate the UMAP components for visualization. Cell type annotation of the clustered groups was performed using cell markers from the CellMarker 2.0 database.

F. Cell Feature Vector

To characterize the biological differences in gene expression profiles across different cell types, we defined cell feature vectors for each type. Using FindMarker, we calculated differential gene expression for each cell type and applied the threshold $|\log FC| > 1$&$p.value < 0.05$ to select marker genes. For cell type A, the marker gene list is denoted as $List_A$. The differential expression of $gene_n$ in cell type A is represented as $\log FC\_n$, leading to the feature vector for cell type A defined as $g_A = [g_1, g_2, \ldots, g_n]$, with the following mapping relation:

$$\mathrm{g_n} = \begin{cases} 1, & \mathrm{n} \in \mathrm{List}_A, \log FC_\mathrm{n} > 0 \\ 0, & \mathrm{n} \notin List_A \\ -1, & \mathrm{n} \in List_A, \log FC_n < 0 \end{cases} \tag{1}$$

G. Calculating Cell Communication by Cellphonedb

To investigate the communication between various cell types and tumor cells in breast cancer, we used the cellphonedb software to analyze cell communication from scRNA-seq data, employing default parameters for the analysis. Communication counts from the results served as an indicator of communication strength between cells.

H. Calculating TF Regulatory Network by Pando

To compute the TF regulatory network from scATAC-seq data, we first calculated the gene activity matrix corresponding to scATAC-seq data using the GeneActivity method from the Signac package and normalized it with LogNormalize, using the median of nCount_RNA as the scale.factor parameter. Subsequently, we computed the TF regulatory network using the Pando package, with motif annotation from Pando's preset version, and the human genome reference sequence version "hg38". The calculated TF regulatory network was organized into an adjacency matrix A, taking into account that multiple genes may be regulated by the same TF. Thus, the adjacency matrix A was transformed as follows:

$$A' = A + A^2 \tag{2}$$

I. Mapping Gene Expression Through Radiomics and the Cancer Genome Atlas (TCGA) Expression Profiles

To align this study with clinical needs, we constructed a deep learning model to integrate radiomics and gene expression data. For the radiomic features F, we built a Decoder to reconstruct expression profile data X:

$$F = g(X) \tag{3}$$

J. Similarity Network Based on GAT

For the single-layer GAT network, the input node features are $h = \overrightarrow{h_1}, \overrightarrow{h_2}, \ldots, \overrightarrow{h_N}, \overrightarrow{h_i} \in \boldsymbol{\mathcal{R}}^F$,where N is the number of nodes, and F is the node feature dimension. The updated features become $h' = \overrightarrow{h'_1}, \overrightarrow{h'_2}, \ldots, \overrightarrow{h'_N}, \overrightarrow{h'_i} \in \boldsymbol{\mathcal{R}}^F$.

The multi-head attention mechanism provides multiple attention representations for each node. For the single attention head i, the attention score $e_{uv}^{(i)}$ between node u and node v can be obtained by calculating the similarity of the features of the two nodes, as given by the formula:

$$e_{uv}^{(i)} = LeakyReLU\left(\alpha^T \left[W\overrightarrow{h_u}||W\overrightarrow{h_v}\right]\right) \tag{4}$$

where $\overrightarrow{h_u}$ is the feature vector for node *u*, W is the trainable weight matrix for transforming node features, and α is the trainable parameter of the attention mechanism.

Attention scores are normalized to ensure that the sum of attention coefficients for all incoming edges of each node equals 1. For node *v*, the normalized attention coefficient α_{uv} is calculated as follows:

$$\alpha_{uv}^{(i)} = \frac{\exp\left(\mathrm{e}_{uv}^{(i)}\right)}{\sum_{k\in \mathrm{N}(v)} \exp(\alpha_{kv}^{(i)})} \tag{5}$$

where N(*v*) is the set of neighbors of node *v*, including node *v* itself, derived from gene homology and protein interaction data.

The updated feature representation of node *v*, ${h'}_v$ is calculated as a weighted sum of the feature representations of its neighboring nodes, with weights given by the attention coefficients:

$${h'}_v{}^{(i)} = \sigma\left(\sum\nolimits_{k\in \mathrm{N}(v)} \alpha_{kv}^{(i)} W^{(i)} h_k\right) \tag{6}$$

where σ is the activation function.

All output features from the heads are concatenated and transformed through a linear layer to produce the final node feature representation:

$$h'_v = W^O \left[{h'_v}^{(1)} || {h'_v}^{(2)} || \ldots || {h'_v}^{(M)} \right] \tag{7}$$

where W^O is a trainable weight matrix used to merge the multi-head representations.

Each base learner is based on GAT. In each base learner, the intercellular communication network between cell types is treated as an adjacency matrix, with the communication strength between each cell type and tumor-related cells serving as the node feature matrix h_c, where cell types are nodes and node attributes indicate whether the communication strength with tumor-related cells exceeds the average.

Additionally, the marker genes of each cell type are organized into the node feature matrix h_g, where genes are nodes and node attributes indicate whether the gene is a marker for a certain cell type. The feature weights for the node features in the communication network are W_c, leading to the weighted node feature matrix h'_g as follows:

$$h'_g = h_g W_c \tag{8}$$

The ensemble learning model averages the outputs of individual base learners to derive the predicted probability of each node's existence in the prior feature gene set, representing its similarity to the prior feature gene set. For an ensemble learning model Hm composed of m base learners, the output is:

$$\bar{p} = \frac{1}{m} \sum\nolimits_{i \in H} p_i \tag{9}$$

where p_i is the predicted probability output by base learner *i*.

K. Prognostic Predicting Network

To predict patient prognosis, we employed a variational Dropout layer to train the dropout rate p'. To enable the discrete dropout process to be differentiable, we relaxed the dropout process using noise perturbation *n*, with the relaxation mask *z* defined as:

$$\begin{aligned} t = {} & log\,(p' + \text{eps}) - \log(1 - p' + eps) \\ & + \log(n + eps) - \log(1 - n + eps) \end{aligned} \tag{10}$$

$$z = 1 - sigmoid\left(\frac{t}{T}\right), T = 0.1 \tag{11}$$

where eps is a small value and $n = random(0, 1)$. The output *h* is the Hadamard product of input *x* and mask $z : h = x * z$ Fitting is performed through a multilayer perceptron, where the l-th layer (*l* = *12*, …,*L*) contains n_l neurons, with weight matrix $W^l_{n_l \times n_{l-1}}$ and bias term $b^l_{n_l \times 1}$, leading to the output of the l-th layer H^l expressed as:

$$H^l = \sigma\left(W^l H^{l-1} + b^l\right) \tag{12}$$

where $H^0 = h$, and H^L is the output of the multilayer perceptron, $H^L = \widehat{h_\theta}\pi(x)$. The model's loss function is LogLikelihood loss:

$$\begin{aligned} l(\theta) = & -\frac{1}{N_{E=1}} \sum_{i:E_i=1} \left(\widehat{h_\theta}(x_i) - \log \sum_{j \in \Re(T_i)} e^{\widehat{h_\theta}(x_j)} \right) \\ & + \lambda \cdot \|\theta\|_2^2 \end{aligned} \tag{13}$$

where *E* is the observable event, and the risk set $\Re(T_i)$ contains patients at time *T* still at risk of failure, with $\widehat{h_\theta}(x)$ being the Hazard function and the output of the multilayer perceptron.

L. Methods Improvement for Parameter Selection and Model Optimization

To determine the optimal hyperparameters and validate the model's performance, a rigorous approach was adopted in this study. The key hyperparameters, including the learning rate, regularization parameters, number of heads in the multi-head attention mechanism, linear layer parameters, Dropout rate *p*, and temperature parameter τ, were selected through systematic optimization (Table S2 and Fig. S4). Below, we detail the methodology used for hyperparameter selection and model validation.

1. Hyperparameter Selection:

a) Learning Rate and Regularization Parameters: The learning rate and regularization parameters are critical for ensuring the model converges to an optimal solution while avoiding overfitting. We employed a grid search approach combined with *k*-fold cross-validation to determine the optimal values. Specifically, we tested a range of learning rates from 0.0001 to 0.01 and regularization parameters (L2 penalty) from 0.00001 to 0.001. The cross-validation process involved partitioning the dataset into *k* = *5* subsets, training the model on *k*-1 subsets, and validating on the remaining subset. This process was repeated *k* times, and the average validation loss was computed to identify the optimal hyperparameters.

The optimization process can be formalized as:

$$\eta^*, \lambda^* = \arg\min_{\eta,\lambda} \frac{1}{k} \sum_{i=1}^{k} L^{(i)}_{\text{val}}(\eta, \lambda) \tag{14}$$

where $L^{(i)}_{\text{val}}$ is the validation loss for the *i*-th fold.

b) Multi-Head Attention Mechanism: The number of heads in the multi-head attention mechanism significantly impacts the model's ability to capture diverse interactions within the data. We evaluated the performance of the model using 2, 4, 6, and 8 attention heads. The optimal number of heads was determined based on the model's performance on the validation set, measured by the area under the receiver operating characteristic curve (AUC-ROC).

The optimal number of heads *H* was determined by:

$$\mathrm{H}^* = \arg\max_{\mathrm{H}} \mathrm{AUC} - \mathrm{ROC}_{\text{val}}(\mathrm{H}) \tag{15}$$

c) Linear Layer Parameters: The parameters of the linear layer, including the number of neurons and weight matrices, were optimized to ensure effective feature transformation and integration. We tested different configurations of the linear layer, ranging from 64 to 512 neurons, and selected the configuration that yielded the highest validation accuracy.

$$\text{Neurons}^* = \arg\max_{\text{Neurons}} \text{Accuracy}_{\text{val}}(\text{Neurons}) \tag{16}$$

d) Dropout Rate p and Temperature Parameter τ: The Dropout rate p and temperature parameter τ were optimized to enhance the model's robustness and generalization capability. The Dropout rate p was varied from 0.1 to 0.5, and the temperature parameter τ was tested in the range of 0.01 to 0.1. The optimal values were determined through cross-validation, ensuring the model achieved the best balance between regularization and information retention.

To predict patient prognosis, we employed a variational Dropout layer to train the dropout rate p. To enable the discrete dropout process to be differentiable, we relaxed the dropout process using noise perturbation n, with the relaxation mask z defined as:

$$z = sigmoid\left(\frac{\log p - \log(1-\mathrm{p}) + \mathrm{n}}{\tau}\right) \tag{17}$$

where τ is a small value and $n \sim Gumbel(0, 1)$. The output h is the Hadamard product of input x and mask z.

The optimal values of p and τ were determined by:

$$p^*, \tau^* = \arg\min_{p,\tau} L_{val}(p, \tau) \tag{18}$$

2. Model Validation:

a) Cross-Validation: To assess the model's generalization performance, we employed k-fold cross-validation. The dataset was divided into $k = 5$ subsets, and the model was trained and validated iteratively on different subsets. The final model performance was reported as the average of the validation metrics across all folds.

b) Performance Metrics The model's performance was evaluated using multiple metrics, including AUC-ROC, F1 score, precision, recall, and accuracy. These metrics provided a comprehensive assessment of the model's predictive capability and robustness (Table S3 and Table S4).

3. Prognostic Prediction Network:

To predict patient prognosis, we used a variational Dropout layer to train the dropout rate p. The discrete dropout process was relaxed using noise perturbation n, with the relaxation mask z defined as:

$$z = sigmoid\left(\frac{\log p - \log(1-\mathrm{p}) + \mathrm{n}}{\tau}\right) \tag{19}$$

where τ is a small value and $n \sim Gumbel(0, 1)$. The output h is the Hadamard product of input x and mask z.

Fitting is performed through a multilayer perceptron (MLP), where the $l - th\ layer\ (l = 1, 2, \ldots, L)$ contains N_l neurons, with weight matrix W_l and bias term b_l, leading to the output of the l -th layer expressed as:

$$h_l = \sigma(W_l h_{l-1} + b_l) \tag{20}$$

where $h_0 = x$, and h_L is the output of the multilayer perceptron.

The model's loss function is the LogLikelihood loss:

$$L = -\sum_{i=1}^{N} \delta_i \log\left(\frac{\exp(-\exp(-\theta_i))}{1 + \exp(-\theta_i)}\right) \tag{21}$$

where δ_i is the observable event, and the risk set R_i contains patients at time T_i still at risk of failure, with θ_i being the Hazard function and the output of the multilayer perceptron.

4. Implementation Details:

All hyperparameters were optimized using the PyTorch library, and the model was trained using the Adam optimizer with a batch size of 64. The training process was monitored using early stopping to prevent overfitting, with a patience of 10 epochs. The final model was trained on the entire dataset using the optimal hyperparameters identified during the validation phase.

By systematically optimizing these hyperparameters and rigorously validating the model, we ensured that our approach achieved superior performance and robustness, as evidenced by the AUC-ROC of 0.839 (95% CI: 0.81–0.87) on the external cohort (TCGA-TNBC). This comprehensive methodology not only enhances the interpretability of our results but also provides a robust framework for future studies in multi-omics data integration.

M. Cytoscape-Guided Biological Findings From MultiGAT Predictions

To derive biologically interpretable findings from the MultiGAT model on the dataset_mine cohort, we integrated model-derived key genes with curated pathway libraries and analyzed the resulting gene–pathway networks in Cytoscape to prioritize candidate genes, pathways, and modules that support prognostic interpretation. The workflow was designed to be reproducible and consistent with the analytical standards used elsewhere in this study.

1) Inputs and candidate gene set: We used the prensemble (Section J–L) to rank genes and formed a candidate set by selecting genes above a predetermined probability threshold (or the top-N ranked list). This set was intersected with robust pathway enrichment results derived from BioCarta 2016 [29], KEGG 2019 [30], and WikiPathways 2019 [31] to ensure biological plausibility and reduce database-specific noise.

2) Gene–pathway bipartite construction: For each library, we constructed a bipartite network whose left part comprises genes and right part comprises pathways. An edge between a gene and a pathway indicates that the gene appears in the enrichment term's gene set for that library. Practically, we built an edge list by ranking terms within each library by the Combined Score and retaining up to the top 30 per library; gene symbols were standardized and trimmed. The consolidated edge list across libraries served as the basis for downstream visualization and topology analysis.

3) Export to Cytoscape and visual encoding: We exported a node table (id, label, type∈{gene, pathway}, degree) and an edge table (source, target, library) for import into Cytoscape. In Cytoscape, we applied a style that maps node size to degree (larger nodes for higher connectivity), node color to type (genes vs pathways), and edge color to a neutral gray with reduced opacity to reduce clutter. Labels were displayed only for high-degree nodes, with a white halo to minimize overlap, and layout followed a grouped/bipartite arrangement to preserve interpretability of gene–pathway relationships.

4) Module detection and prioritization: We identified network modules using community detection (e.g., Markov clustering or Louvain via clusterMaker2) and prioritized modules by:

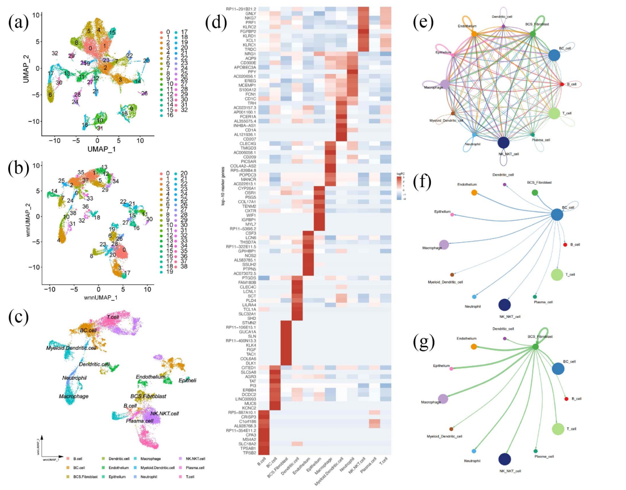

Fig. 2. Dimensionality Reduction, Clustering, Cell Annotation and Cell Communication Networks. (a) Dimensionality reduction of scRNA-seq data visualized via UMAP, showcasing the clustering of cells based on gene expression profiles alone (n = 3 independent experiments). (b) Dimensionality reduction of scRNA and CITE-seq data visualized via UMAP, incorporating both RNA-seq and protein surface marker data to improve cellular clustering and visualization (n = 3 independent experiments). (c) Annotation of cell types based on marker gene expression in RNA+CITE-seq clusters, where distinct cell populations were identified through the integration of both transcriptomic and proteomic data. Each cluster was annotated based on the expression of well-known marker genes (n = 3 independent experiments). (d) Heatmap of marker gene expression for each cell type, displaying the differential expression of key marker genes across the identified cell clusters in RNA+CITE-seq data (n = 3 independent experiments). (e) Total cell communication network derived from analysis of multiple cell types within the TME, showcasing the complex intercellular interactions across the entire cell population (n = 3 independent experiments). (f) Cancer-associated cell subnetwork of BC cells, highlighting key interactions between breast cancer (BC) cells and various tumor-associated cells, including fibroblasts, immune cells, and endothelial cells (n = 3 independent experiments). (g) Cancer-associated cell subnetwork of BCS Fibroblasts, illustrating the interactions between breast cancer stem cells (BCSCs) and the stromal components of the TME, particularly cancer-associated fibroblasts (n = 3 independent experiments).

(i) within-module gene and pathway degrees (hubness), (ii) cross-library recurrence (genes and pathways present in ≥ 2 libraries), and (iii) consistency with MultiGAT interpretability (high SHAP embedding contributions or elevated attention on relevant edges). Modules satisfying these criteria were considered biologically credible and robust across knowledge sources.

(5) Cross-modal and clinical consistency checks: We integrated ligand–receptor correlation analyses (Section G and downstream LR correlation pipeline) with module membership to assess whether highly connected genes were positively associated with clinically relevant ligand–receptor signaling. When a ligand–receptor pair showed significant correlation with multiple module genes, we treated this as supporting evidence for a mechanistic chain linking intercellular communication to downstream pathway activation. Consistency with TCGA-derived expression and radiomics mappings further strengthened the prognostic relevance.

(6) Reporting and figure generation: We summarized module-level findings by exporting Cytoscape figures and annotated networks, alongside programmatically generated faceted heatmaps of gene–pathway presence to improve legibility in the manuscript. The resulting visualizations emphasize cross-library consensus, highlight hub genes and recurrent pathways, and directly support biological claims about the TME and pathway activation in TNBC.

This integrative network analysis bridges MultiGAT predictions with established pathway knowledge, providing an interpretable, library-aware framework for identifying robust candidate genes and mechanisms. By combining network topology, cross-library recurrence, model interpretability signals (attention and SHAP), and ligand–receptor correlation, the approach yields biologically coherent findings aligned with prognostic modeling objectives.

III. Results

A. *Integrated Single-Cell Profiling Reveals Enhanced Cellular Heterogeneity*

To dissect the TME of TNBC, we integrated scRNA-seq and CITE-seq data from the GSE199219 dataset [12], [13]. After data preprocessing, the dataset comprised 59426 cells with 12 genes retained for analysis2. Uniform Manifold Approximation and Projection (UMAP) clustering of RNA-seq data alone resolved 32 distinct cellular subpopulations Fig. 2(a). However, incorporating CITE-seq-derived protein surface markers increased resolution to 38 subpopulations Fig. 2(b), highlighting its capacity to uncover phenotypic diversity obscured by transcriptomics alone. Annotation of RNA+CITE-seq clusters using canonical marker genes identified critical TME components, including breast cancer (BC) cells, cancer-associated fibroblasts (BCS Fibroblasts), and immune subsets Fig. 2(c) [14]. Differential gene expression analysis further defined molecular signatures unique to each cluster Fig. 2(d), revealing BC cell-specific pathways linked to therapy resistance and stromal activation.

B. *Cell-Cell Communication Networks Drive Tumor-Stromal Crosstalk*

Intercellular communication analysis using CellphoneDB uncovered a dense network of ligand-receptor interactions across the TME Fig. 2(e) [15]. BC cells exhibited robust signaling crosstalk with fibroblasts (e.g., TGF-β, CXCL12-CXCR4), endothelial cells (e.g., EGF-EGFR), and myeloid subsets (e.g., CSF1-CSF1R) Fig. 2(f). Notably, BCS Fibroblasts emerged as central hubs, coordinating stromal remodeling via interactions with macrophages (IL6-IL6R) and neutrophils (SEMA4D-PLXNB2) Fig. 2(g). These findings implicate fibroblast-driven signaling as a key mediator of immune suppression and extracellular matrix remodeling in TNBC [17], [18].

C. *Chromatin Accessibility Maps Implicate Master Regulators of Malignancy*

After data preprocessing, the dataset included 50520 samples and 2209 genes. Analysis of scATAC-seq datasets (GSE168026 and GSE212707) resolved chromatin landscapes across TNBC subclones [3], [17], [18], [19], [20], [21], [22], [23]. UMAP visualization highlighted epigenetically distinct cell states Fig. 3(a), with regulatory network reconstruction identifying FOXM1, STAT3, and NF-κB as master regulators of malignant chromatin

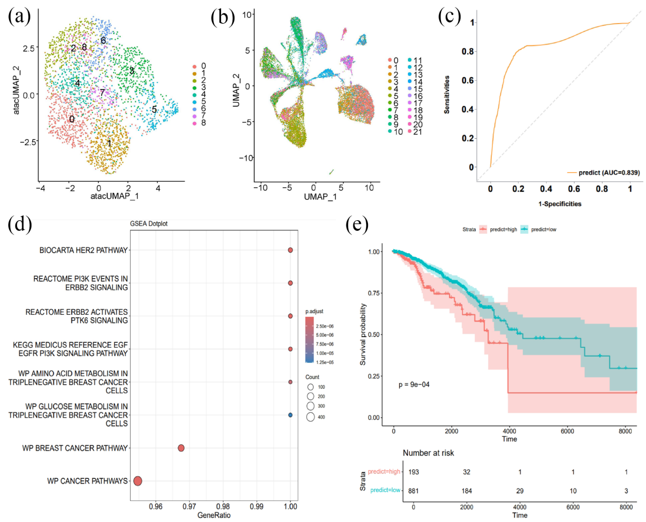

Fig. 3. Multi-dimensional Analysis and Prognostic Prediction of Key Genes in TNBC Using ATAC-seq and GAT Model. (a) Dimensionality reduction of GSE168026 ATAC-seq data visualized via UMAP (n = 3 independent experiments). (b) Dimensionality reduction of GSE212707 ATAC-seq data visualized via UMAP (n = 3 independent experiments). (c) Receiver operating characteristic (ROC) curve for GAT model prediction, evaluating the performance of the GAT in predicting key prognostic genes (n = 3 independent experiments). (d) Pathway enrichment analysis of predicted genes, illustrating the biological pathways and molecular processes significantly enriched within the set of genes identified by the GAT model (n = 3 independent experiments). (e) Kaplan-Meier (KM) survival plot for prognostic prediction, showing the correlation between the expression levels of the predicted key genes and patient survival outcomes (n = 3 independent experiments).

in GSE168026. Subtype-specific transcription factor hubs, such as RUNX1 in basal-like TNBC (GSE212707), linked chromatin dynamics to transcriptional programs driving proliferation and immune evasion Fig. 3(b) and Fig. S2.

D. GAT-Based Similarity Network Predicts Prognostic Biomarkers

We constructed a gene similarity network using a GAT that integrated multi-omics features, gene homology, PPI networks (comprising 707236 PPIs involving 17753 genes, with each edge representing two protein IDs), and TF regulatory edges [20], [21], [22], [23]. The model achieved an AUC-ROC of 0.839 (95% CI: 0.81–0.87) Fig. 3(c), prioritizing 48 high-confidence TNBC-associated genes (e.g., *PI3K, ERBB2, PTK6, EGF/EGFR* and *PD-L1*). Pathway enrichment linked these genes to cell cycle dysregulation ($p = 1.2\times10^{-7}$, PI3K-Akt signaling ($p = 3.8\times10^{-5}$, and immune checkpoint pathways Fig. 3(d). A prognostic MLP model trained on these genes stratified TCGA-TNBC patients into high- and low-risk cohorts with divergent survival outcomes (log-rank $p = 0.002$; Fig. 3(e)), validating their clinical relevance.

E. Cytoscape-Guided Network Analysis Identifies Cross-Library Consensus Modules

To translate MultiGAT-derived predictions into biologically coherent mechanisms, we constructed and analyzed a gene–pathway bipartite network across BioCarta 2016 [29], KEGG 2019 [30], and WikiPathways 2019 [31] using Cytoscape-compatible exports. Programmatic extraction from the consolidated edge list yielded 42194 gene–pathway edges linking 1665 genes to 90 pathways (library composition: KEGG_2019_Human = 21687 edges; WikiPathways_2019_Human = 16911 edges; BioCarta_2016 = 3596 edges). Network hub analysis revealed MAPK1, MAPK3, AKT1, PIK3R1/PIK3CA, HRAS, RAF1, and MAP2K1/2 as top-degree genes, while "Pathways in cancer", "PI3K-Akt signaling pathway", "MAPK signaling pathway", "Cytokine–cytokine receptor interaction", and "Neuroactive ligand–receptor interaction" ranked among the most connected pathways (Fig. S8). Cross-library recurrence analysis highlighted genes present in all three libraries (e.g., MAPK3, MAPK8/12/13/14, ZYX, CLIP1), supporting robust mechanistic relevance beyond any single database.

Consistent with model interpretability (Fig. S5, Fig. S6 and Fig. S7), SHAP-derived modality contributions on dataset_mine indicated strong ATAC2 and PPI influences (ATAC2, shap_total_abs = 0.1198; PPI, 0.1176; homolog, 0.0873; ATAC1, 0.0399), while attention-weight aggregation favored ATAC2 edges (alpha_mean: ATAC2 = 0.702; ATAC1 = 0.178; homolog = 0.126; PPI = 0.078). These signals converge on TF-centric regulatory connectivity (ATAC2) and canonical interaction networks (PPI), aligning with the observed dominance of PI3K-Akt/MAPK axis and cytokine signaling in the Cytoscape network. Together, the cross-library consensus of hubs and the modality-specific interpretability profiles suggest that TNBC biofindings are driven by coordinated signaling modules encompassing PI3K-Akt/MAPK cascades, focal adhesion/ECM dynamics, and cytokine-mediated immunomodulatory pathways, coherent with the prognostic stratification results above.

IV. Discussion

TNBC is an aggressive subtype characterized by the absence of ER, PR, and HER2 expression, limiting therapeutic options and contributing to poor prognosis [24], [25], [26], [27], [28]. Despite advances in multi-omics profiling, its high molecular heterogeneity and lack of actionable biomarkers remain major clinical challenges [34], [35], [36], [37], [38], [39]. In this study, we addressed these gaps by integrating scRNA-seq, CITE-seq, scATAC-seq, and radiomics data via a GAT, a graph-based deep learning approach designed to capture cross-modal interactions overlooked by single-omics methods (Table S5 and Fig. S9) [33], [35]. Our analysis uncovered profound heterogeneity within the TME of TNBC, combining scRNA-seq and CITE-seq identified 38 distinct cellular subpopulations (vs. 32 from scRNA-seq alone), including cancer-associated fibroblasts (BCS fibroblasts), immune subsets, and breast cancer (BC) cells [1], [14], [15]. CellphoneDB analysis revealed dense ligand-receptor networks, with BC cells engaging in robust crosstalk with stromal and immune components, and BCS fibroblasts acting as key signaling hubs mediating extracellular matrix remodeling and immune suppression [2], reinforcing the stromal compartment's critical role in TNBC progression [12].

Further insights from scATAC-seq highlighted epigenetically distinct TNBC subclones, with FOXM1, STAT3, NF-κB, and RUNX1 identified as master regulators of chromatin accessibility programs linked to proliferation and immune escape [3], [17], [23]. The GAT model (AUC-ROC = 0.839) validated well-characterized oncogenic pathways (PI3K-Akt, EGFR/ErbB, PD-L1-associated signaling) [25], [35], serving as internal confirmation of its biological fidelity [37]. Beyond these canonical pathways, our integrative analysis uncovered novel regulatory axes, complement and coagulation signaling (linking innate immunity and vascular function) [15], [25], ECM-integrin-focal adhesion pathways (governing immune accessibility via mechanotransduction) [1], [22], leukocyte transendothelial migration (vascular gatekeeping for immune trafficking) [2], sphingolipid signaling (metabolic-immune coupling) [21], and nanoparticle-mediated receptor activation (nano-bio interface modulation) [23]. These findings reframe TNBC biology from a tumor-intrinsic, gene-centric paradigm to a multi-scale model centered on TME interface interactions [21], [35].

From a clinical perspective, these discoveries offer actionable implications. Targeting stromal hubs, vascular gatekeeping mechanisms, or metabolic-immune loops (e.g., integrin $\alpha6\beta4$ or SPHK1 inhibition) could enhance immune cell infiltration and immunotherapy efficacy [18], [21]. The prognostic gene signatures (both established and novel) and epigenetic regulators identified enable precision diagnostic tools and individualized risk stratification [24], [36]. Additionally, the GAT-based multi-omics integration framework is scalable, holding promise for future clinical decision support systems across cancer types [22], [27]. Collectively, these advances address the unmet need for targeted therapies in TNBC by expanding the repertoire of potential therapeutic targets and biomarkers [28], [36]. This study highlights the power of integrating multi-omics data with graph-based deep learning to unravel the complex biological architecture of TNBC. By identifying both well-established oncogenic drivers and previously underexplored, TME-centric pathways, our framework provides a more comprehensive view of TNBC progression. Moving beyond the limitations of single-omics analyses, this approach opens new avenues for robust biomarker discovery and therapeutic innovation. Importantly, by reframing TNBC through the lens of multi-scale tumor-microenvironment interactions, this work not only advances mechanistic understanding but also lays a foundation for personalized medicine strategies aimed at improving outcomes in this clinically challenging breast cancer subtype.

V. Conclusion

In summary, this study presents a comprehensive multi-omics approach to understanding the TME of TNBC. By integrating single-cell RNA-seq, CITE-seq, and scATAC-seq data with a graph-based deep learning model MultiGAT, we identified novel cellular interactions, master regulatory networks, and key prognostic biomarkers, including both well-characterized oncogenic drivers and understudied pathways linked to innate immune-vascular crosstalk, mechanotransduction, metabolic-immune coupling, and nano-bio interactions. Our findings reveal the multi-scale complexity of the TME and highlight the critical role of stromal-immune- vascular interactions in tumor progression. The use of GAT models to integrate diverse omics data provided a powerful framework for predicting clinically relevant biomarkers, which could inform the development of personalized therapies for TNBC. This integrated approach, which transcends the limitations of single-omics studies, offers new opportunities for biomarker discovery and therapeutic innovation in TNBC, potentially improving patient outcomes and advancing precision medicine.

V. Conflict of Interest

The authors declare that they have no conflict of interest.

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